

Department of ECE / EEE
Jan–May 2026
UE24EE241B–CONTROLSYSTEMS
A/H/P (Group project) – Guidelines

1.	Faculty will identify project teams consisting of four students.
2.	Each team will be working on the transfer function of a system provided by the faculty.
3.	Teams are expected to carry out the following activities on an ongoing basis throughout Unit 4
	Activities
	a) Stability Analysis: Examine whether the selected plant is stable by observing pole-zero location in s-plane. Draw p-z plot using MATLAB '<i>p-z map</i>'. b) Time Response Analysis: Excite the plant with a unit step input and obtain the response plot. Comment on the relevant transient response parameters (as applicable) and steady-state characteristics based on the obtained plot. c) Steady-State Error Analysis: Consider the plant as part of a unity negative feedback control system. Identify the system TYPE and compute the static error constants and steady-state errors for unit step, unit ramp, and unit parabolic inputs. d) Gain and Stability Study: Introduce a gain 'K' in cascade with the plant in the unity feedback configuration. Derive the characteristic equation of the system. Apply the Routh–Hurwitz stability criterion and determine the range of 'K' for which the closed-loop system is: (a) stable, (b) marginally stable, and (c) unstable. e) Simulation and Verification: Using MATLAB/Simulink, simulate the system described above for a unit step input. Choose appropriate values of 'K' corresponding to stable, marginally stable, and unstable conditions. Verify whether the simulation results support the analytical conclusions. f) Root Locus Analysis: Obtain the root locus of the loop transfer function analytically and determine the stability range of 'K'. Compare these results with those obtained earlier. Use the MATLAB command '<i>rlocus</i>' to plot the root locus and compare it with the analytical plot. Present your observations.

	g) Bode Plot Analysis: Sketch the Bode plot of the loop transfer function analytically and compute the gain margin and phase margin. Comment on the closed-loop stability. Verify the results using the MATLAB 'bode' command and compare the margins obtained. State your inferences. h) Nyquist Stability Analysis: Draw the Nyquist plot of the loop transfer function analytically and determine the gain and phase margins. Comment on the closed-loop stability. Verify the plot using the MATLAB 'nyquist' command and compare the results. Summarize your observations. i) Controller Design: Design controllers such that the unity feedback system achieves a peak overshoot of less than 10% and a settling time of less than 2 seconds. Design and analyze the following controllers: (a) Proportional (P), (b) Proportional–Integral (PI), (c) Proportional–Derivative (PD), and (d) Proportional–Integral–Derivative (PID). Simulate the system response for each controller and comment on their performance. Provide justification for the chosen controller gains. <i>Note: Use of the PID Toolbox for simulation is not permitted.</i>
5	Teams are encouraged to utilize MATLAB/Simulink for analysis and simulation.
6	Upon completion of the project, each team shall submit a spiral-bound written / typed report, followed by a 10-minute PowerPoint presentation for final evaluation scheduled during 14th week of CoE. The report must mandatorily include the MATLAB/ Simulink codes.

Date: 21-01-2026

Anchor: Dr. Anuradha M

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